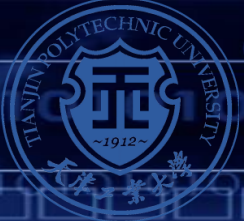




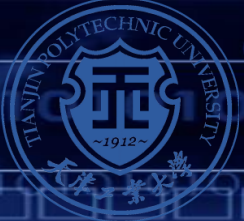
实验5 gre配置



目 标

❖ 学完本章内容后，同学们将能够：

- 了解GRE技术原理
- 掌握防火墙中GRE配置流程



本章目录

1

GRE 基本原理回顾

2

实验拓扑设计与搭建

3

实验网络配置

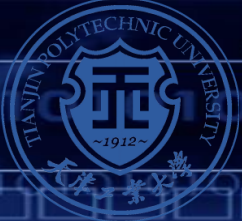
4

GRE配置



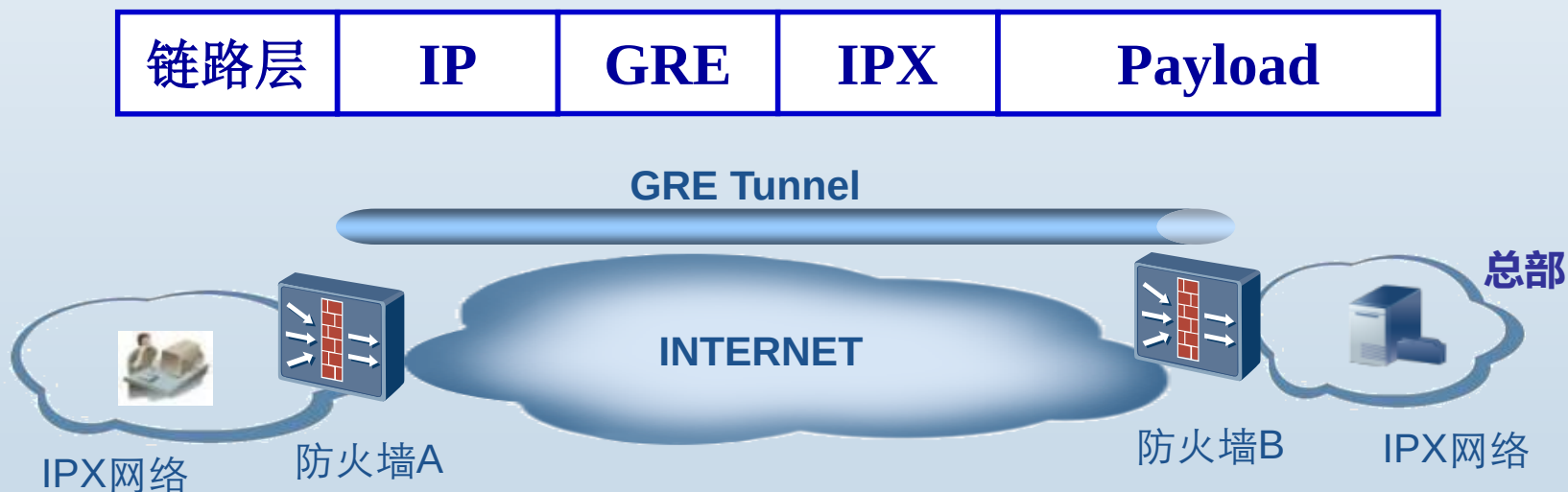
1

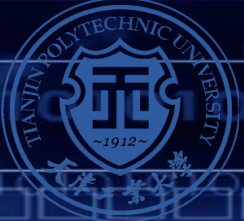
GRE原理



GRE协议概述

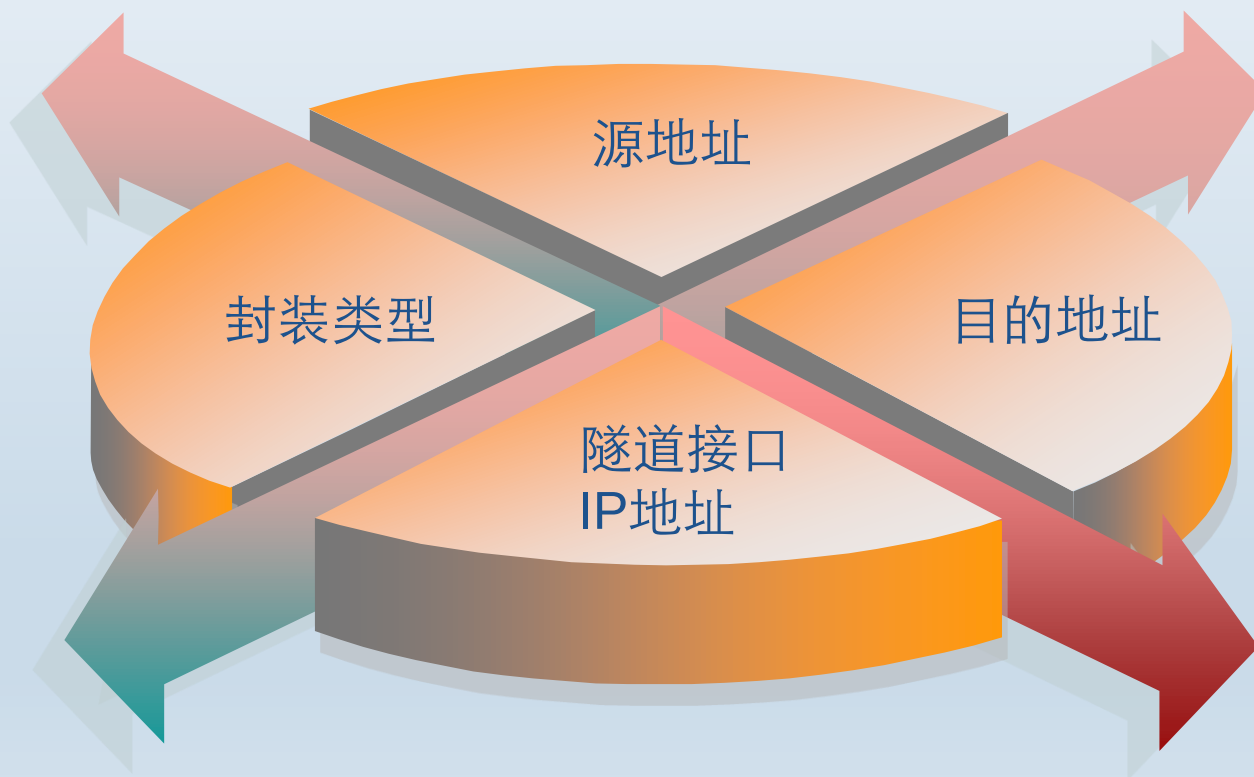
- ❖ GRE (Generic Routing Encapsulation): 是对某些网络层协议（如：IP, IPX, AppleTalk等）的数据报进行封装，使这些被封装的数据报能够在另一个网络层协议（如IP）中传输

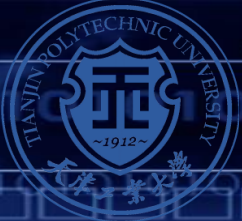




GRE的实现-隧道接口

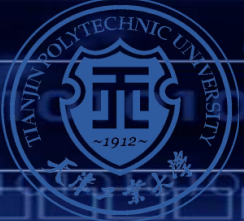
- ❖ 隧道接口（Tunnel接口）是为实现报文的封装而提供的一种点对点类型的虚拟接口，与Loopback接口类似，都是一种逻辑接口





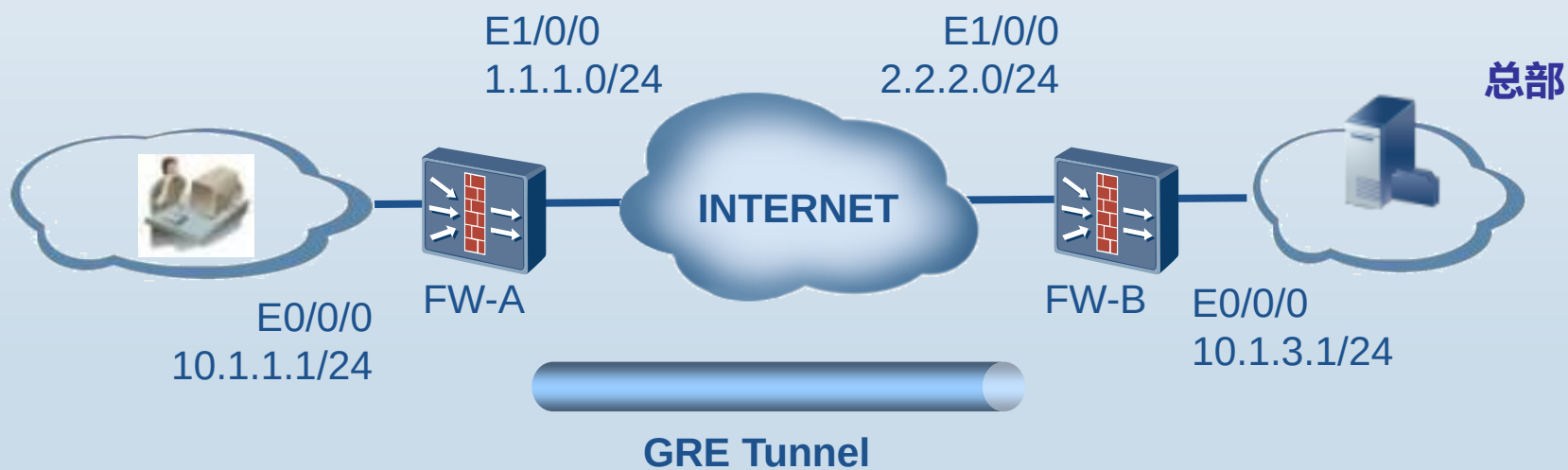
GRE的实现-封装与解封装





GRE VPN典型应用场景描述

- ❖ FW-A和FW-B通过Internet相连，两者公网路由可达。网络1和网络2是两个私有的IP网络，通过在两台FW之间建立GRE隧道实现两个私有IP网络互联。

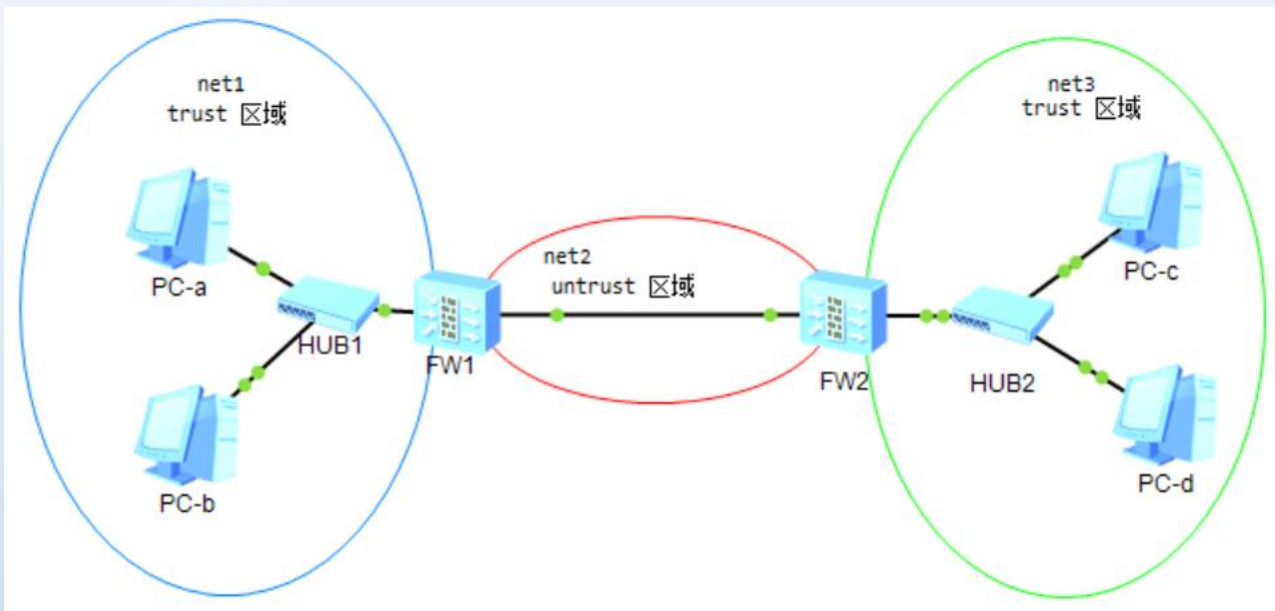




2

实验规划

设备与拓扑



实验设备：
USG6000: 2台
HUB: 2台
PC: 4台

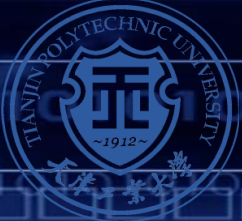


数据规划

网络设备名称	接口	IP地址	所属安全区域	互联对端设备	对端接口
FW1	G1/0/0	2.x.y.1/24	untrust	FW2	G1/0/0
	G1/0/1	1.x.y.254/24	trust	HUB1	e0/0/0
FW2	G1/0/0	2.x.y.2/24	untrust	FW2	G1/0/0
	G1/0/1	3.x.y.254/24	trust	HUB2	e0/0/0
HUB1	e0/0/0	无	无	FW1	G1/0/1
	e0/0/1	无	无	PC-a	
	e0/0/2	无	无	PC-b	
HUB2	e0/0/0	无	无	FW2	
	e0/0/1	无	无	PC-c	
	e0/0/2	无	无	PC-d	

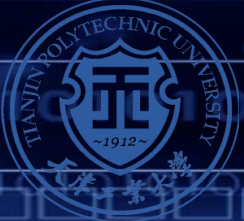
主机名称	IP地址	子网掩码	默认网关
PC-a	1.x.y.1	255.255.255.0	1.x.y.254
PC-b	1.x.y.2	255.255.255.0	1.x.y.254
PC-c	3.x.y.1	255.255.255.0	3.x.y.254
PC-d	3.x.y.2	255.255.255.0	3.x.y.254

特别注意：IP地址中的x,y的含义
x：学号倒数第三位； y:学号倒数后两位



接口规划：FW1

设备名称	项目	数据	说明
防火墙 FW1	Tunnel 1	IP地址：10.x.y.1/24 安全区域：dmz 封装协议：tunnel-protocol gre 封装后的源IP：source 2.x.y.1 封装后的目的IP：destination 2.x.y.2 GRE封装关键字验证：gre key cipher 12345	GRE 封装隧道接口：通过在隧道接口中配置封装参数，对内网报文进行相应的GRE封装
	指定路由	目的地址：3.x.y.0 下一跳：Tunnel1	配置路由，将需要经过GRE隧道传输的流量引入到GRE隧道中。



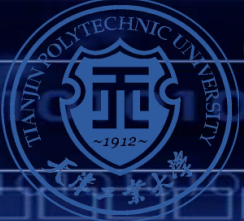
接口规划：FW2

设备名称	项目	数据	说明
防火墙 FW2	Tunnel 1	IP地址：10.x.y.2/24 安全区域：dmz 封装协议：tunnel-protocol gre 封装后的源IP：source 2.x.y.2 封装后的目的IP：destination 2.x.y.1 GRE封装关键字验证：gre key cipher 12345	GRE 封装隧道接口：通过在隧道接口中配置封装参数，对内网报文进行相应的GRE封装
	指定路由	目的地址：1.x.y.0 下一跳：Tunnel1	配置路由，将需要经过GRE隧道传输的流量引入到GRE隧道中。



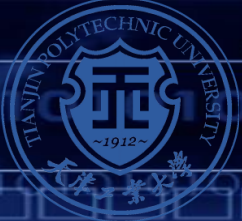
3

实验步骤



实验步骤

- ❖ 构建拓扑
- ❖ 启动设备
- ❖ 配置PC的IP地址和默认网关
- ❖ 配置防火墙
- ❖ 结果验证



配置PC

PC-a

基础配置 命令行 组播 UDP发包工具 串口

主机名: PC-a

MAC 地址: 54-89-98-24-78-2D

IPv4 配置

☒ 静态 ☐ DHCP ☐ 自动获取 DNS 服务器地址

IP 地址: 1 . 1 . 1 . 1

子网掩码: 255 . 255 . 255 . 0

网关: 1 . 1 . 1 . 254

DNS1: 0 . 0 . 0 . 0

DNS2: 0 . 0 . 0 . 0

IPv6 配置

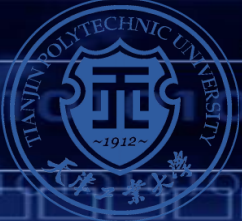
☒ 静态 ☐ DHCPv6

IPv6 地址: ::

前缀长度: 128

IPv6 网关: ::

应用



启动设备-首次登录

- ❖ 防火墙首次启动登录需要验证用户身份
- ❖ 采用系统初始默认用户名和密码
- ❖ 首次登录后需要立即修改密码
- ❖ 修改为统一密码，方便日后重复调试

```
FW1
FW1
Username:
Error: The username times out.
Username:admin
Password:
Error: Authentication fail

Username:admin
Password:
The password needs to be changed. Change now? [Y/N]: y
Please enter old password:
Please enter new password:
Please confirm new password:

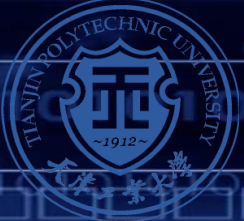
Info: Your password has been changed. Save the change to survive
*****
*           Copyright (C) 2014-2018 Huawei Technologies Co., Ltd.
*           All rights reserved.
*           Without the owner's prior written consent,
*           no decompiling or reverse-engineering shall be allowed.
*****

<USG6000V1>
```



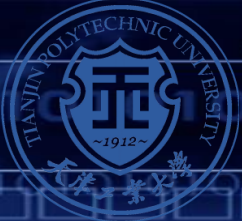
设备首次开启

- ❖ 默认用户名：admin
- ❖ 默认密码：Admin@123
- ❖ 设置新密码：admin@123



华为防火墙命令行模式

- ❖ **sysname**: 配置设备名称
- ❖ **Interface g0/0/0**: 进入接口配置模式
- ❖ **display** : 用于在任何视图下显示相关内容
- ❖ **Ctrl+z** 快捷方式, 退出到<> 用户模式
- ❖ **save**: 在用户视图模式下完成配置文件的保存
- ❖ **display ip routing-table**: 显示路由表
- ❖ **display zone**: 显示区域
- ❖ **display firewall session table**: 显示防火墙会话表
- ❖ **display security-policy all (自定义名称)** : 显示安全策略



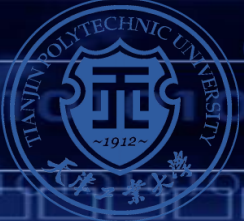
GRE VPN配置思路

基础配置

配置tunnel
逻辑接口

配置到对端
网络内网网
段的路由

放开相应的
域间规则



配置防火墙FW1

❖ 修改设备名称

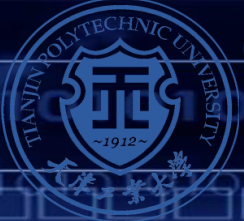
- sysname 学号后三位-1

❖ 配置缺省安全策略

- security-policy
- default action deny

❖ 清空缺省路由

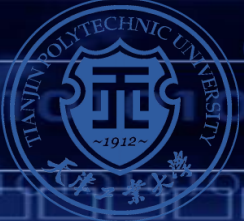
undo ip route-static 0.0.0.0 0.0.0.0 2.2.2.2



FW1配置- 接口

❖ 配置接口地址

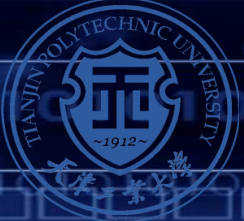
- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.1 255.255.255.0(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 1.x.y.254 255.255.255.0
- service-manage ping permit



FW1配置- tunnel接口

❖ 配置tunnel接口

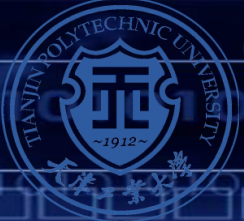
- interface Tunnel1 （创建隧道接口1）
- ip address 10.x.y.1 255.255.255.0 （配置隧道接口地址）
- tunnel-protocol gre （配置隧道封装协议类型）
- source 2.x.y.1 （封装后的源IP）
- destination 2.x.y.2 （封装后的目的IP）
- gre key cipher 12345 （gre安全验证关键字）



FW1配置-安全区域

❖ 配置接口地址FW-A上的安全区域并添加相应接口:

- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0
- firewall zone dmz
- add interface Tunnel1



配置防火墙FW2

❖ 修改设备名称

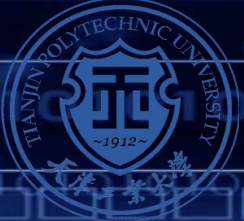
- sysname 学号后三位-2

❖ 配置缺省安全策略

- security-policy
- default action deny

❖ 清空缺省路由

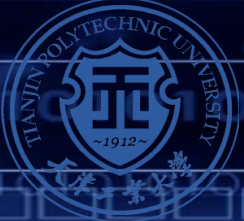
undo ip route-static 0.0.0.0 0.0.0.0 2.2.2.1



FW2配置- 接口

❖ 配置接口地址

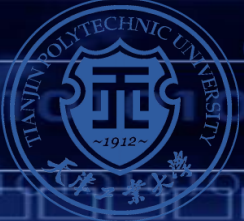
- interface GigabitEthernet1/0/0 (配置接口G1/0/0)
- ip address 2.x.y.2 255.255.255.0(配置IP地址)
- service-manage ping permit (开启ping 服务)
- interface GigabitEthernet1/0/1
- ip address 3.x.y.254 255.255.255.0
- service-manage ping permit



FW2配置- tunnel接口

❖ 配置tunnel接口

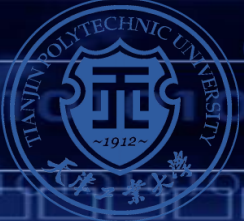
- interface Tunnel1 （创建隧道接口1）
- ip address 10.x.y.2 255.255.255.0 （配置隧道接口地址）
- tunnel-protocol gre （配置隧道封装协议类型）
- source 2.x.y.2 （封装后的源IP）
- destination 2.x.y.1 （封装后的目的IP）
- gre key cipher 12345 （gre安全验证关键字）



FW2配置-安全区域

❖ 配置接口地址FW-A上的安全区域并添加相应接口:

- firewall zone trust
- add interface GigabitEthernet1/0/1
- firewall zone untrust
- add interface GigabitEthernet1/0/0
- firewall zone dmz
- add interface Tunnel1



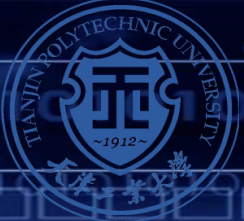
FW1配置-静态路由

❖ 配置静态路由

ip route-static 3.x.y.0 255.255.255.0 Tunnel1

目的私网网络地址

本地tunnel接口



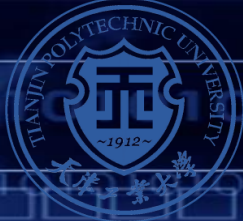
FW2配置-静态路由

❖ 配置静态路由

ip route-static 1.x.y.0 255.255.255.0 Tunnel1

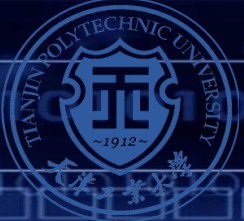
↓
目的私网网络地址

↓
本地tunnel接口



FW1和FW2的安全策略匹配条件

业务方向	设备	源安全区域	目的安全区域	源地址	目的地址
PC-A访问PC-c	FW1	local	untrust	2.x.y.1/32	2.x.y.2/32
		trust	dmz	1.x.y.0/24	3.x.y.0/24
	FW2	Untrust	local	2.x.y.2/32	2.x.y.1/32
		dmz	Trust	3.x.y.0/24	1.x.y.0/24
PC-c访问PC-A	FW	Untrust	local	2.x.y.1/32	2.x.y.2/32
		dmz	Trust	1.x.y.0/24	3.x.y.0/24
	FW-b	local	untrust	2.x.y.2/32	2.x.y.1/32
		trust	dmz	3.x.y.0/24	1.x.y.0/24



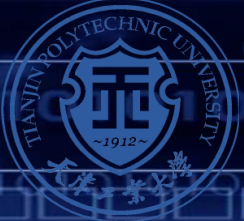
FW1配置-安全策略

❖ 策略1：允许trust与dmz区主机互相访问

```
security-policy  
rule name t-d  
source-zone trust  
source-zone dmz  
destination-zone trust  
destination-zone dmz  
action permit
```

❖ 策略2：允许untrust与local区主机互相访问

```
security-policy  
rule name u-l  
source-zone local  
source-zone untrust  
destination-zone local  
destination-zone untrust  
service gre  
action permit
```



FW2配置-安全策略

❖ 策略1：允许trust与dmz区主机互相访问

- security-policy
- rule name t-d
- source-zone trust
- source-zone dmz
- destination-zone trust
- destination-zone dmz
- action permit

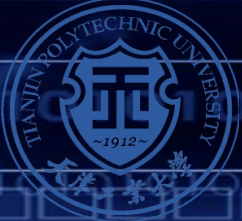
❖ 策略2：允许untrust与local区主机互相访问

- security-policy
- rule name u-l
- source-zone local
- source-zone untrust
- destination-zone local
- destination-zone untrust
- service gre
- action permit



3

结果验证



结果验证

❖ 查看接口信息

```
[test-1]display ip interface brief
```

```
2023-04-23 04:40:20.680
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
(d): Dampening Suppressed
```

```
(E): E-Trunk down
```

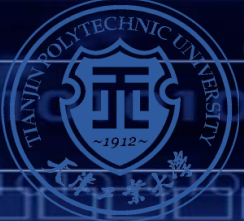
```
The number of interface that is UP in Physical is 5
```

```
The number of interface that is DOWN in Physical is 6
```

```
The number of interface that is UP in Protocol is 5
```

```
The number of interface that is DOWN in Protocol is 6
```

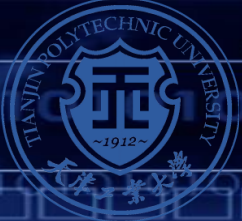
Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	192.168.0.1/24	down	down
GigabitEthernet1/0/0	2.2.2.1/24	up	up
GigabitEthernet1/0/1	1.1.1.254/24	up	up
GigabitEthernet1/0/2	unassigned	down	down
GigabitEthernet1/0/3	unassigned	down	down
GigabitEthernet1/0/4	unassigned	down	down
GigabitEthernet1/0/5	unassigned	down	down
GigabitEthernet1/0/6	unassigned	down	down
NULL0	unassigned	up	up (s)
Tunnel1	10.1.1.1/24	up	up
Virtual-if0	unassigned	up	up (s)



结果验证

❖ 查看安全区域

```
[test-1]display zone interface
2023-04-23 04:41:49.880
local
  interface of the zone is (0):
#
trust
  interface of the zone is (2):
    GigabitEthernet0/0/0
    GigabitEthernet1/0/1
#
untrust
  interface of the zone is (1):
    GigabitEthernet1/0/0
#
dmz
  interface of the zone is (1):
    Tunnel1
#
```



结果验证-FW1

❖ 查看路由表

```
[test-1]display ip routing-table
```

```
2023-04-23 04:44:36.360
```

```
Route Flags: R - relay, D - download to fib
```

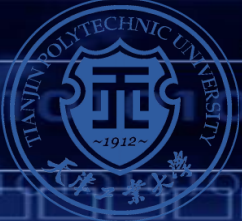
特别提示：不允许出现0.0.0.0 0.0.0.0 2.2.2.2 这样的记录，否则为零分

```
-----  
Routing Tables: Public
```

```
Destinations : 9
```

```
Routes : 9
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.0/24	Direct	0	0	D	1.1.1.254	GigabitEthernet
1/0/1						
1.1.1.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
1/0/1						
2.2.2.0/24	Direct	0	0	D	2.2.2.1	GigabitEthernet
1/0/0						
2.2.2.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
1/0/0						
3.3.3.0/24	Static	60	0	D	10.1.1.1	Tunnel1
10.1.1.0/24	Direct	0	0	D	10.1.1.1	Tunnel1
10.1.1.1/32	Direct	0	0	D	127.0.0.1	Tunnel1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0



结果验证-FW1

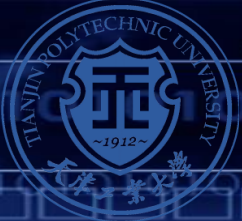
❖ 查看安全策略

```
<test-1> display security-policy rule all
```

```
2023-04-23 05:34:06.730
```

```
Total:3
```

RULE ID	RULE NAME	STATE	ACTION	HITS
1	t-d	enable	permit	0
2	u-l	enable	permit	0
0	default	enable	deny	0

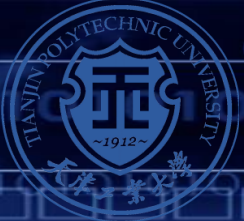


验证网络连通性

❖ 在PC-a上分别pingPC-d的IP地址

The screenshot shows a PC simulator window titled "PC-a". It has several tabs: "基础配置", "命令行" (highlighted with a red box), "组播", "UDP发包工具", and "串口". The "命令行" tab is active, displaying a terminal window with the following text:

```
Welcome to use PC Simulator!  
PC>ping 3.3.3.2  
Ping 3.3.3.2: 32 data bytes, Press Ctrl C to break  
From 3.3.3.2: bytes=32 seq=1 ttl=126 time=16 ms  
From 3.3.3.2: bytes=32 seq=2 ttl=126 time=16 ms  
From 3.3.3.2: bytes=32 seq=3 ttl=126 time<1 ms  
From 3.3.3.2: bytes=32 seq=4 ttl=126 time=15 ms  
From 3.3.3.2: bytes=32 seq=5 ttl=126 time=16 ms  
--- 3.3.3.2 ping statistics ---  
5 packet(s) transmitted  
5 packet(s) received  
0.00% packet loss  
round-trip min/avg/max = 0/12/16 ms
```



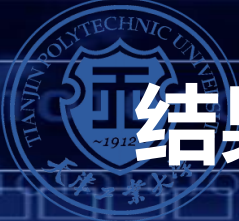
结果验证-FW1查看会话表

❖ 在PC-a上使用Ping命令后，立即查看FW1上的会话表

```
[test-1]display firewall session table verbose
2023-04-23 04:51:27.830
Current Total Sessions : 3
icmp VPN: public --> public ID: c387fe4778cd7109b46444b934
Zone: trust --> dmz TTL: 00:00:20 Left: 00:00:02
Recv Interface: GigabitEthernet1/0/1
Interface: Tunnell NextHop: 3.3.3.2 MAC: 0000-0000-0000
<--packets: 1 bytes: 60 --> packets: 1 bytes: 60
1.1.1.1:26551 --> 3.3.3.2:2048 PolicyName: t-d

icmp VPN: public --> public ID: c487fe4778cd6381db36444b933
Zone: trust --> dmz TTL: 00:00:20 Left: 00:00:01
Recv Interface: GigabitEthernet1/0/1
Interface: Tunnell NextHop: 3.3.3.2 MAC: 0000-0000-0000
<--packets: 1 bytes: 60 --> packets: 1 bytes: 60
1.1.1.1:26295 --> 3.3.3.2:2048 PolicyName: t-d

gre VPN: public --> public ID: c487fe4778cd3b032346444b930
Zone: untrust --> local TTL: 00:10:00 Left: 00:09:42
Recv Interface: GigabitEthernet1/0/0
Interface: InLoopBack0 NextHop: 127.0.0.1 MAC: 0000-0000-0000
<--packets: 0 bytes: 0 --> packets: 5 bytes: 440
2.2.2.2:0 --> 2.2.2.1:0 PolicyName: u-1
```



结果验证-FW1的G1/0/0接口抓包结果

```
> Frame 9: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: HuaweiTe_f7:72:40 (00:e0:fc:f7:72:40), Dst: HuaweiTe_a2:04:df (00:e0:fc:a2:04:df)
  0 Internet Protocol Version 4, Src: 2.2.2.2, Dst: 2.2.2.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 88
    Identification: 0x0052 (82)
  > Flags: 0x00
    Fragment Offset: 0
    Time to live: 255
    Protocol: Generic Routing Encapsulation (47)
    Header Checksum: 0xb31e [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 2.2.2.2

  0 Generic Routing Encapsulation (IP)
    0 Flags and Version: 0x2000
      0... .. = Checksum Bit: No
      .0.. .. = Routing Bit: No
      ..1. .... = Key Bit: Yes
      ...0 .... = Sequence Number Bit: No
      .... 0... .. = Strict Source Route Bit: No
      .... .000 .... = Recursion control: 0
      .... .. 0000 0... = Flags (Reserved): 0
      .... .. .. .000 = Version: GRE (0)
      Protocol Type: IP (0x0800)
      Key: 0x00003039
```

验证位

十进制" 12345 "



END